PATENT QUALITY EVALUATION

Patent Value Explorer

Scores patents against the OECD Patent Quality framework, normalized to technology-field and filing-year cohorts.

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#supportbyheart 20+ year patent & accessibility expertise

The problem

Patent quality is hard to compare

Raw patent metrics are not comparable across technology fields or filing periods. Ten forward citations in biotechnology mean something different than ten in mechanical engineering. A 2002 patent has had two decades to be cited; a 2022 patent has barely started.

Manual expert assessment is slow and expensive. It does not scale to portfolios of dozens or hundreds of patents.

What is needed: a fast, reproducible first-pass screen, anchored to a published framework rather than ad-hoc heuristics.

Overview

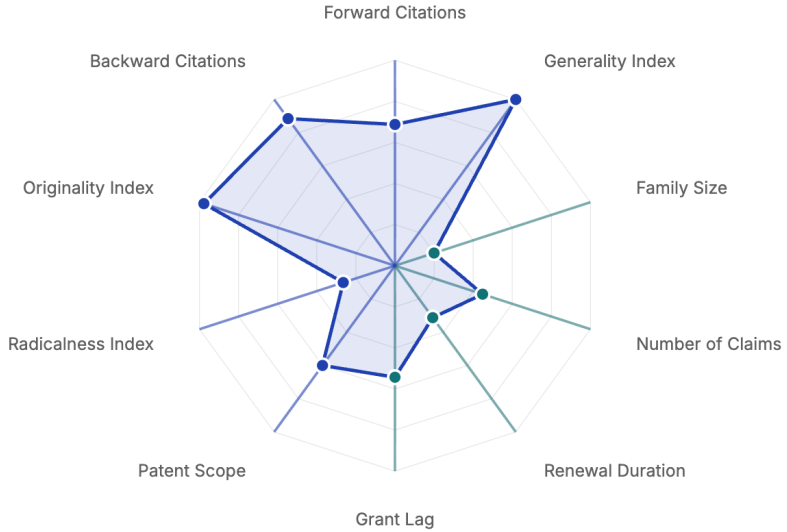
What Patent Value Explorer does

Input

A patent publication number in any common format (EP1234567, US-9,876,543, WO2020/123456, with or without kind codes).

Output

- Normalized score (0.0-1.0) for each of 10 OECD indicators
- Cohort percentile and qualitative interpretation per indicator
- Composite Quality Index (mean of 5 OECD components)
- Radar chart grouped into Technological Importance and Market Relevance
- Breakthrough or Highly-Cited badge when applicable
- Optional AI narrative summary (Anthropic Claude)



Compared to 28.990 patents in Control filed in 2007

Technological Importance · Market Relevance · PVE

Built on the OECD Patent Quality framework

Source: Squicciarini, Dernis & Criscuolo (2013), *"Measuring Patent Quality: Indicators of Technological and Economic Value"*, OECD Science, Technology and Industry Working Papers 2013/03.

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OECD concepts implemented

10

scored indicators + Breakthrough flag

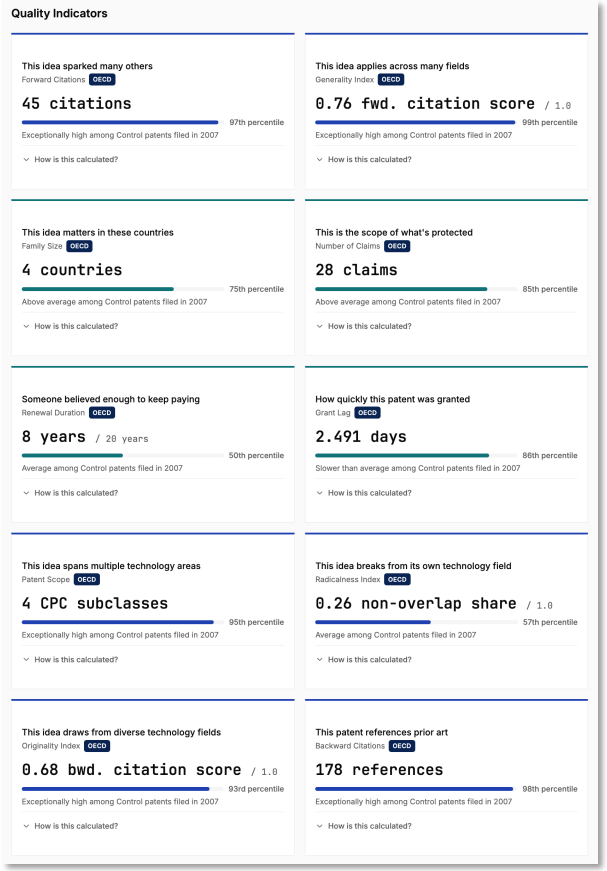
5

components in the composite index

Two OECD concepts are not covered: citations to non-patent literature (NPL), which requires a dataset not available in BigQuery PATSTAT, and the X/I/Y-only variant of Forward Citations, which the authors describe as essentially identical to standard forward citations.

The ten OECD indicators

#	Indicator	Measures
1	**Forward Citations**	Technological impact, how many later families cite this patent
2	**Backward Citations**	Knowledge base breadth, how many prior-art references are cited
3	**Patent Scope**	Technological breadth, distinct CPC subclasses on the patent
4	**Family Size**	International market relevance, jurisdictions in the DOCDB family
5	**Generality Index**	Cross-field applicability, diversity of citing patents' CPC sections
6	**Originality Index**	Breadth of knowledge sources, diversity of cited patents' CPC sections
7	**Radicalness Index**	Share of backward citations in CPC subclasses outside the focal patent
8	**Grant Lag**	Examination speed, days from filing to grant
9	**Number of Claims**	Scope of legal protection, number of patent claims
10	**Renewal Duration**	Sustained commercial value, maximum renewal fee year paid



Plus a Breakthrough Invention flag, awarded when the forward-citation percentile reaches the top 1% of the cohort.

Cohort normalization is the core mechanism

Each indicator is normalized against the patent's primary WIPO technology field and filing-year cohort. That is what makes scores comparable across fields and across decades.

1

Cohort lookup

Retrieve the pre-computed percentile distribution (p1, p5, p25, p50, p75, p95, p99) for the (field, year, indicator) triple.

2

Winsorization

Clamp the raw value to the p1 to p99 range to suppress outliers.

3

Linear scaling

Map the winsorized value to the 0.0 to 1.0 range using the cohort's p1 and p99.

4

Percentile interpolation

Compute the patent's exact percentile inside the cohort by linear interpolation between stored breakpoints.

16,348 cohorts 35 WIPO technology fields, 47 filing years (1978 to 2024), 10 indicators. Sparse cohorts are omitted; the dataset ships as static JSON.

How the composite is built

The Composite Quality Index

Equal-weighted mean of five OECD composite components:

Composite = mean(Forward Citations, Family Size, Number of Claims, Originality, Radicalness)

Notes

These five components can be computed at request time without prohibitive cost.

Generality is excluded from the standard composite. It requires a ~16 GB per-patent BigQuery scan, so it is offered on-demand via a button in the UI.

Backward Citations, Patent Scope, Grant Lag, and Renewal Duration are shown as standalone indicators. Per the OECD framework, they are not part of the composite.

When a component is missing (e.g. patent not yet granted), the composite uses the remaining components and the count is shown, for example "based on 4 of 5 components."

Layered on top of the OECD baseline

Everything OECD-defined carries an OECD badge in the UI. Everything added by PVE carries a PVE badge, so the boundary between framework and interpretation stays visible.

Citation-rank title badges

Top 1% gets "Breakthrough" (amber). Top 10% gets "Highly Cited" (blue). Below the threshold, no badge; the percentile stays on the Forward Citations card.

Patent Archetype

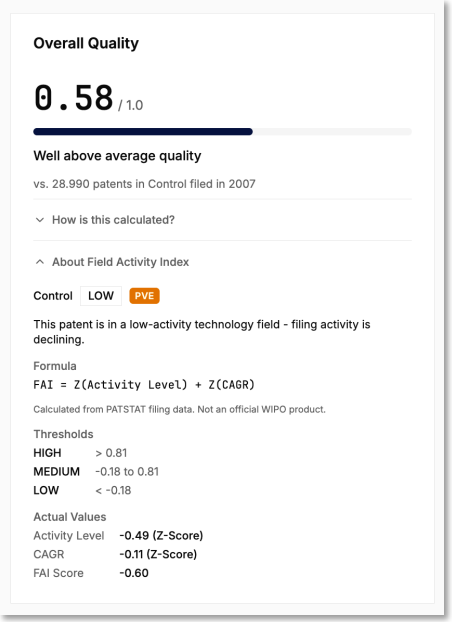
Specialist, Generalist, Disruptor, or Incremental. Derived from the indicator profile. Used as a deterministic fallback when the AI narrative is unavailable.

Field Activity Index (FAI)

Per-field activity classification (HIGH, MEDIUM, LOW), CAGR, and a composite Z(activity) + Z(CAGR). Adapted from the WIPO Patent Momentum Indicator.

EPO dimension grouping

Indicators are visually grouped into Technological Importance and Market Relevance on the radar chart, mirroring the OECD paper's framing.



How a request becomes a profile

Architecture and data flow

Patent number

Parsed and normalized

3-tier lookup

Reference JSON, then in-memory cache, then fresh PATSTAT

9 parallel queries

Promise.allSettled, isolated failures

Cohort normalization

Static JSON, no runtime DB

Composite + display

Radar, cards, optional narrative

Tier	Storage	Latency
Reference patents	Static JSON shipped with the app (14 curated patents)	< 100 ms
Patent-result cache	In-memory Map (Node process)	< 10 ms
Cohort statistics	Static JSON, 16,348 cohorts	< 50 ms
Fresh lookup	PATSTAT MCP server to BigQuery (9 parallel queries)	10 to 25 s
Runs inside EPO TIP via a one-click Jupyter notebook launcher, or as any standard Node 20+ app.		

How it is built

Tech stack and code quality

Stack

- SvelteKit 2 with adapter-node
- Svelte 5 (Runes) for fine-grained reactivity
- TypeScript 5.9, strict mode across the codebase
- Tailwind CSS 4 and bits-ui (shadcn-svelte)
- LayerChart 2 (D3-based) for the radar visualization
- Valibot for runtime schema validation
- PATSTAT access via MCP server to BigQuery
- Anthropic Claude API for the optional AI narrative

Quality

- 231 unit tests (vitest) covering indicators, normalization, composite, formatting
- JSDoc on scoring functions references the OECD section they implement
- No any in scoring logic; shared types in src/lib/scoring/types.ts
- Valibot validates route parameters and MCP responses at the boundary
- ESLint and Prettier configured
- Reproducible builds via committed package-lock.json (npm ci)
- Graceful degradation: indicator failures are isolated; AI narrative is non-fatal

WCAG 2.1 AA target

Accessibility

Keyboard navigation	Arrow-key cycling through radar axes. Enter to search, Escape to clear.
Screen-reader support	Semantic roles (role="search", role="application", role="meter"), ARIA labels on interactive elements, and text alternatives where applicable.
Bar-chart alternative	Tabular bar chart with role="meter" and aria-valuenow / aria-valuemin / aria-valuemax on each bar. Used on small viewports and exposed to assistive technology.
Colour contrast	EPO-dimension colour pairs (blue and teal) meet at least 3:1 against both light and dark backgrounds.
Motion safety	Tailwind motion-safe: prefix and @media (prefers-reduced-motion: reduce) suppress animations system-wide.

THANKS

Patent Value Explorer

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Repository

github.com/mtcberlin/epo-tip-patent-value-explorer

Organisation

Moving Targets Consulting, mtc.berlin